

TRANSPORTATION TECHNOLOGY

Transportation affects our lives in a multitude of ways. Transportation systems move raw materials to manufacturers and finished products to consumers locally, nationally, and globally. Individuals use transportation systems every day for business, work, and pleasure.

Transportation technology courses provide students with opportunities to understand transportation systems from the perspective of either the consumer or the service provider. The range of courses enables students to study both vehicle ownership and vehicle maintenance, and to develop skills and prepare for careers in the servicing and repair of vehicles, aircraft, and/or watercraft. In addition, students will develop the Essential Skills and work habits that are important for success in the transportation industry.

The list of approved emphasis areas for transportation technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Transportation Technology: Vehicle Maintenance, Grade 12

Workplace Preparation

TTJ4E

This course introduces students to the servicing, repair, and maintenance of vehicles through practical applications. The course is appropriate for all students as a general interest course to prepare them for future vehicle operation, care, and maintenance or for entry into an apprenticeship in the motive power trades. Students will develop an awareness of environmental and societal issues related to transportation, and will learn about careers in the transportation industry and the skills and training required for them.

Prerequisite: None

A. VEHICLE MAINTENANCE FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate a working knowledge of the components and principles of operation of a vehicle powertrain and the procedures involved in servicing it;
- A2.** demonstrate an understanding of the interior and exterior care and maintenance of a vehicle;
- A3.** demonstrate a working knowledge of the technical requirements, procedures, tools and equipment, and documentation connected with servicing and maintaining a vehicle or small-engine product.

SPECIFIC EXPECTATIONS

A1. Vehicle Powertrains

By the end of this course, students will:

- A1.1** identify the components of an internal combustion engine and describe its operation and the routine service required to keep it operating at peak efficiency;
- A1.2** explain the similarities and differences among various engine cycles (*e.g., two-stroke, four-stroke, rotary, turbine*);
- A1.3** explain powertrain terminology commonly used in the transportation industry (*e.g., double overhead camshaft [DOHC], automatic transmission [A/T], all-wheel drive [AWD]*);
- A1.4** identify the components of a vehicle drivetrain (*e.g., constant velocity [C/V] joints and boots, axles, chain or belt drive, power take-off [PTO], transmission*) and describe the operation of a drivetrain and the routine service required to keep it operating at peak efficiency.

A2. Interior and Exterior Care and Maintenance

By the end of this course, students will:

- A2.1** describe, on the basis of research, a variety of interior vehicle cleaning products and their applications;
- A2.2** identify the types of polishes, waxes, and cleaners that are appropriate for various vehicle finishes (*e.g., exterior base clear coat, gel coat*);

A2.3 describe various repair techniques and make minor repairs to a vehicle body finish (*e.g., stone-chip repair, small-dent repair*);

A2.4 identify and describe various types of fastening methods (*e.g., welding, nut and bolt, riveting, chemical bonding*) used in maintenance procedures for vehicles and small-engine products;

A2.5 identify the consequences and legal implications of various vehicle customizations (*e.g., ground effects, body alterations, audio systems, light intensity, window tinting*).

A3. Service and Maintenance

By the end of this course, students will:

- A3.1** locate appropriate information sources (*e.g., shop manuals, owner's manual, online databases*) and consult as required for specifications, tools, equipment, and procedures used in servicing and maintaining vehicles or small-engine products;
- A3.2** locate, use, and correctly interpret assembly drawings that depict the components of vehicle systems or small-engine products;
- A3.3** identify and describe the function of common hand tools (*e.g., wrenches, socket/ratchet set, screwdrivers*), power tools (*e.g., electric drill, die grinder, air ratchet*), and equipment (*e.g., battery charger, hoist, parts washer*) used in servicing and maintaining a vehicle or small-engine product;
- A3.4** identify and describe the function of common measuring tools (*e.g., micrometer, hydrometer, multimeter*) used in servicing and maintaining a vehicle or small-engine product;

A3.5 perform mathematical calculations related to servicing and maintaining a vehicle or small-engine product (*e.g., calculate quantities, ratio of water to antifreeze, fuel consumption*), using appropriate resources (*e.g., owner's manual, service information*);

A3.6 report on work in progress and work performed (*e.g., by completing a work order, parts list, service record, and/or journal*), using appropriate terminology specific to vehicles or small-engine products (*e.g., acronyms, service terms*).

B. VEHICLE MAINTENANCE SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** perform general service and maintenance on vehicles or small-engine products safely, using the owner's manual, repair manuals, tool and equipment manuals, and identification and information labels;
- B2.** inspect, test, and service powertrain components, using appropriate service repair resources;
- B3.** demonstrate a basic understanding of body, brake, steering, and suspension systems and components, and procedures required for their maintenance and service;
- B4.** develop appropriate solutions to various vehicle repair problems or challenges.

SPECIFIC EXPECTATIONS

B1. Service Information

By the end of this course, students will:

- B1.1** identify the meaning of the letters and numbers of the vehicle identification number (VIN) (*e.g., place of origin, engine size, production year*) on the basis of their placement in the VIN;
- B1.2** locate information in the owner's manual and on the vehicle or small-engine product (*e.g., safety warnings, warning light and icon information, computer trouble codes, specifications such as tire size and recommended pressure, identification labels, graphics*) and apply as required when performing service and maintenance procedures;
- B1.3** identify procedures recommended in the owner's manual and/or repair manual for emergency situations (*e.g., flat tire, dead battery, vehicle breakdown*);
- B1.4** demonstrate the correct use of hand, power, machine, and pneumatic tools and equipment required for service tasks (*e.g., tire machine, floor jacks and hoists, safety stands, shop tools*), store them safely, and maintain them in good working order;
- B1.5** remove and replace components (*e.g., engine oil, tires*) correctly, using appropriate product information and specifications (*e.g., engine oil – viscosity and quantity; tires – tire size and load rating*) as noted in the owner's manual and/or repair manual;

- B1.6** demonstrate the safe operation of a variety of heating, cutting, and welding equipment in performing service and maintenance tasks.

B2. Powertrain Systems

By the end of this course, students will:

- B2.1** perform service procedures on an engine lubrication system (*e.g., change engine oil, change oil filter*) safely and correctly;
- B2.2** perform inspection and service procedures on an engine fuel system (*e.g., change fuel filter*) safely and correctly;
- B2.3** perform inspection, testing, and service procedures on an engine cooling system (*e.g., perform pressure test, test freezing/boiling point*) safely and correctly;
- B2.4** perform inspection and service procedures on an engine ignition system (*e.g., replace spark plugs, replace ignition wires*) safely and correctly;
- B2.5** perform inspection, testing, and service procedures on an electrical system (*e.g., test fuses, charge battery, inspect alternator*) safely and correctly;
- B2.6** identify drivetrain components (*e.g., transmission dipstick, differential fill plug, transfer case level plug*) and perform drivetrain maintenance and service (*e.g., clutch adjustment, fluid level checks*) safely and correctly;

- B2.7** perform inspection and service procedures on an engine exhaust system (*e.g., replace a muffler*) safely and correctly.

B3. Body, Brake, Steering, and Suspension Systems

By the end of this course, students will:

- B3.1** locate and identify the major components of body, brake, steering, and suspension systems (*e.g., fender, emergency brake, pneumatic air brake components, tires, hydraulic steering and brake components, shocks*);
- B3.2** correctly interpret assembly drawings that depict the components of body, brake, steering, and suspension systems;
- B3.3** perform maintenance and service procedures related to body, brake, steering, and suspension systems (*e.g., body: hood or door adjustment, paint or finish maintenance; brakes: disc and drum brake service, air brake service; steering: tire and wheel service, hydraulic component service, repacking trailer bearings; suspension: strut and shock inspection*);
- B3.4** repair minor damage to body components (*e.g., panels, lights*) as required to maintain a vehicle in good condition.

B4. Repair Problems or Challenges

By the end of this course, students will:

- B4.1** demonstrate the use of an appropriate diagnostic problem-solving process (*e.g., use of flow charts*) to solve a repair problem (*e.g., no-start condition, no-charging condition*);
- B4.2** design and fabricate a transportation-related project (*e.g., a welding/metal fabrication project*) in response to a challenge, using appropriate tools, equipment, and processes;
- B4.3** report on the repair process or project and its results, and identify possible further repair/service or improvements after completing and testing the repair solution or project.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of ways in which various aspects of the transportation industry affect the environment, and ways in which harmful effects can be remedied or reduced;
- C2.** demonstrate an understanding of the relationship between various aspects of the transportation industry and society.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** demonstrate an understanding of ways in which the transportation industry affects the environment and of efforts being made to remedy or reduce harmful effects (*e.g., improved production methods, automotive parts recycling*), including ways of disposing of waste products (*e.g., used oil, used batteries, used paints/thinners*);
- C1.2** explain the pros and cons of using environmentally friendly products (*e.g., biodegradable cleaners*) and procedures (*e.g., recycling of antifreeze*) in the repair and service of vehicles or small-engine products;
- C1.3** describe appropriate actions to be taken in the event of a spill of waste products (*e.g., used oil, antifreeze, fuel*), and demonstrate the ability to safely implement such actions (*e.g., outline the steps described in an emergency action plan and carry them out*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** research and report on the development of improved safety features in today's vehicles or small-engine products (*e.g., airbags, anti-lock brakes, electronic stability control, back-up alarms, engine kill switch, no-mow-in-reverse switch*);
- C2.2** assess the pros and cons of various types of fuel/energy sources (*e.g., gasoline, propane, diesel, electrical/battery power, biodiesel, hybrid powerplant, hydrogen power cells*) used to power vehicles or small-engine products, taking into account a variety of perspectives (*e.g., consumer's perspective: cost to purchase, cost to operate, performance, emission levels; service/repair perspective: training, safety issues, new tools/equipment required*);
- C2.3** explain how current trends in transportation technology (*e.g., extended maintenance schedules, improved emission standards and testing, use of high-tech components, emphasis on fuel efficiency, manufacturers' efforts to increase the amount of recyclable material in vehicles and small-engine products*) are related to societal attitudes and behaviour.

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and compliance with occupational health and safety regulations and standards;
- D2.** identify and describe career opportunities in the transportation industry and the skills and training required for entry into these careers.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** demonstrate an understanding of legislation and regulations related to procedures and operations used in transportation technology facilities (e.g., *Occupational Health and Safety Act [OHSA]; regulations and standards outlined in the Workplace Hazardous Materials Information System [WHMIS]; Apprenticeship and Certification Act [ACA]*);
- D1.2** demonstrate good housekeeping and safety practices in the work environment (e.g., *cleaning up spills and leaks, keeping areas clean and clear of obstruction*);
- D1.3** use protective clothing and equipment (e.g., *eye protection, breathing apparatus*) as required to ensure their own and others' safety in the work environment;
- D1.4** describe and demonstrate the ability to follow safety precautions applicable to new technologies when working around or servicing vehicles (e.g., *precautions regarding high current and voltage, capacitor discharge rate of supplemental restraint systems, extreme temperature of exhaust systems*);
- D1.5** identify potential health risks when servicing vehicles and demonstrate the use of safe procedures to mitigate these hazards (e.g., *preventive measures to minimize airborne particles and fumes*).

D2. Career Opportunities

By the end of this course, students will:

- D2.1** describe a variety of career opportunities in the transportation industry and the training required for entry into these careers;
- D2.2** demonstrate an understanding of and apply the Essential Skills that are important for success in the transportation industry, as identified in the Ontario Skills Passport (e.g., *problem solving, finding information, measurement and calculation*);
- D2.3** demonstrate an understanding of and apply the work habits that are important for success in the transportation industry, as identified in the Ontario Skills Passport (e.g., *working safely, teamwork, reliability*);
- D2.4** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in transportation technology (e.g., *Passport to Safety certificate, skills checklist, Ontario Skills Passport Work Plan and Transition Plan*), and explain why having a current portfolio is important for career development and advancement.